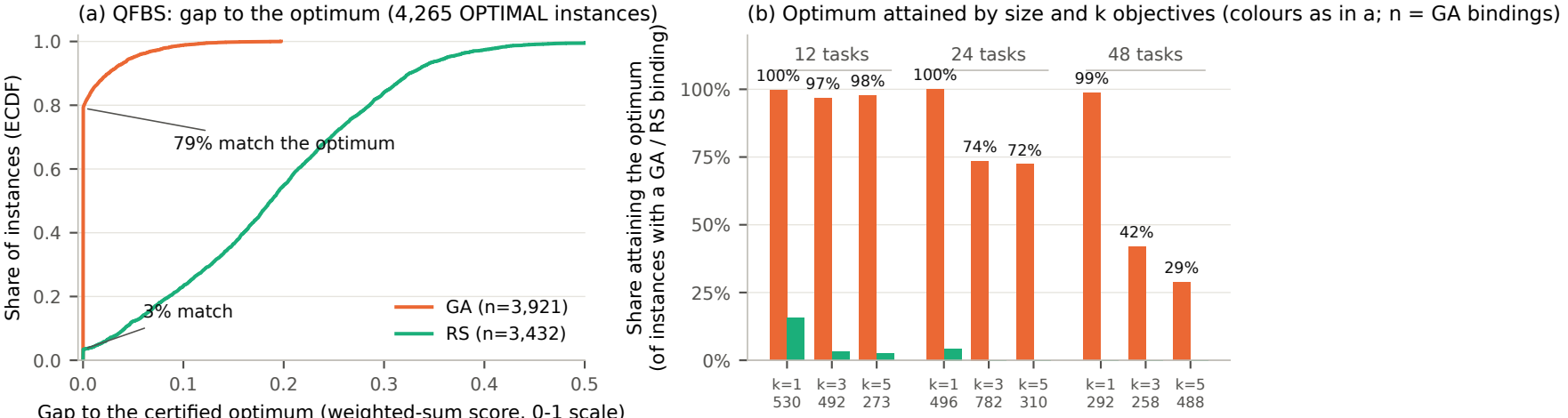
(a) QFBS: gap to the optimum (4,265 OPTIMAL instances)
Share of instances (ECDF)
1.0
0.8
0.6
0.4
0.2
0.0
79% match the optimum
3% match
GA (n=3,921)
RS (n=3,432)
0.0
0.1
0.2
0.3
0.4
0.5
Gap to the certified optimum (weighted-sum score, 0-1 scale)
(b) Optimum attained by size and k objectives (colours as in a; n = GA bindings)
Share attaining the optimum
(of instances with a GA / RS binding)
12 tasks
24 tasks
48 tasks
100%
97%
98%
100%
74%
72%
99%
42%
29%
100%
75%
50%
25%
0%
k=1
530
k=3
492
k=5
273
k=1
496
k=3
782
k=5
310
k=1
292
k=3
258
k=5
488